

Physics, 2018 EC, entrance exam, version 5

Grade 12 · Physics · 2018 E.C.

Time allowed: 120 minutes

Questions: 54

Total marks: 54

Name _____ Grade _____ ID No. _____ Date _____

1. A scientist conducts an experiment to record the boiling point of water at various altitudes. Which of the following statements best justifies the type of knowledge gained from this experiment? (1)
 - A. It is memory-based knowledge because the scientist recalls past experiences
 - B. It is testimonial knowledge because the scientist relies on previous published data
 - C. It is experiential knowledge because the scientist directly observes the phenomena
 - D. It is introspective knowledge because the scientist reflects on their own observations
2. Which of the following best explains why physics is important for chemistry? (1)
 - A. Physics replaces chemical experiments with mathematical models only
 - B. Physics and chemistry study unrelated aspects of matter without overlap
 - C. Physics provides laws governing matter and energy for chemical reactions
 - D. Physics focuses solely on elements and ignores molecules in chemical systems
3. How does Pascal's principle assist soft-bodied animals like earthworms? (1)
 - A. Pressure gradients create electrical signals in neurons
 - B. Pressure in fluid transmits evenly to produce movement
 - C. Pressure difference remove excess water from the body
 - D. Pressure in muscles generates heat for energy production
4. During a military air-defense mission, a RADAR station scans the sky to detect an approaching aircraft. How does it identify the distant target? (1)
 - A. It analyzes sound waves emitted from moving targets
 - B. It measures temperature changes produced by the target
 - C. It sends electromagnetic waves and receives reflected echoes
 - D. It detects gravitational attraction from far away objects remotely
5. A vector has magnitude V . When the angle with the horizontal increases from 30° to 60° , what happens to the horizontal component? (1)
 - A. It becomes half
 - B. It remains the same
 - C. It becomes $\sqrt{3}$ times original value
 - D. It becomes $1/\sqrt{3}$ times original value
6. Two vectors are added using both the triangle and parallelogram methods. Which conclusion is correct? (1)
 - A. Only the triangle method can be applied to vectors at an angle
 - B. Both methods give the same resultant in magnitude and direction
 - C. The parallelogram method ignores the direction of the vectors completely
 - D. The triangle method always gives a larger resultant magnitude than the other
7. The unit vector in the direction of vector $A = 5\mathbf{i} - 12\mathbf{j}$ is (1)
 - A. $(\mathbf{i} - \mathbf{j})/\sqrt{13}$
 - B. $13(\mathbf{i} + \mathbf{j})/13$
 - C. $(5\mathbf{i} + 12\mathbf{j})/13$
 - D. $(5\mathbf{i} - 12\mathbf{j})/13$

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8. Work done by a constant force F for displacement d is given by $W = F \cdot d$. A particle moves along a displacement $(5\mathbf{i} + 5\sqrt{3}\mathbf{k})$ m under a force $(10\mathbf{j} - \sqrt{3}\mathbf{k})$ N. What is the work done by the force? (1)
- A. 35 J
B. -15 J
C. $5\sqrt{3}$ J
D. -50 J
-
9. A train accelerates from rest to 30 m/s in 60 seconds. Four students make statements about the motion: (1)
- Student P: 'The average velocity is 15 m/s, and the instantaneous velocity at 60 s is 30 m/s.' Student Q: 'The average velocity is 30 m/s, and the instantaneous velocity at 60 s is 15 m/s.' Student R: 'Both the average and instantaneous velocity are 15 m/s.' Student S: 'Both the average and instantaneous velocities are 30 m/s.' Which student's statement is correct?
- A. Student R
B. Student S
C. Student P
D. Student Q
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10. Two particles start from the same point at the same time. Particle A starts from rest and accelerates uniformly at 2 m/s^2 , while particle B moves with a constant velocity of 10 m/s. After how many seconds the two particles will be at the same position again, other than $t = 0$? (1)
- A. 10 s
B. 12 s
C. 5 s
D. 8 s
-
11. A vehicle accelerates uniformly from time t_1 to t_2 , moves with constant velocity from time t_2 to t_3 , and decelerates uniformly from t_3 to t_4 . In which time interval are instantaneous and average acceleration both zero? (1)
- A. Between t_1 to t_2
B. Between t_2 to t_3
C. Between t_3 to t_4
D. Between t_1 to t_4
-
12. A ball is thrown vertically upward from the top of a 15 m high tower with an initial speed of 10 m/s. Neglecting air resistance, how long does it take for the ball to reach the ground? (1)
- A. 3 s
B. 4 s
C. 5 s
D. 2 s
-
13. Two projectiles are launched with equal speeds at complementary angles (θ and $90^\circ - \theta$), where the two launch angles are not equal. A student concludes that their maximum heights must be equal. How should this conclusion be judged? (1)
- A. Incorrect, because acceleration at the peaks differs
B. Correct, because both projectiles have equal ranges
C. Correct, because horizontal velocity components are equal
D. Incorrect, because maximum height depends on vertical velocity
-
14. A ball is projected at certain angle with initial velocity u , covers horizontal range R . With what initial velocity it should be projected keeping the angle of projection the same so that its horizontal range becomes $2.25R$? (1)
- A. $2.5u$
B. $1.5u$
C. $2.25u$
D. $0.25u$

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15. A rigid disc rotates about its center. If one particle on the rim sweeps an angular displacement of θ radians, what can be concluded about a particle located halfway between the center and the rim during the same time interval? (1)
- A. It sweeps a smaller angular displacement and travels a longer arc length
 - B. It sweeps the same angular displacement but travels a shorter arc length
 - C. It sweeps a smaller angular displacement but travels the same arc length
 - D. It sweeps the same angular displacement and travels the same arc length
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16. A wheel initially at rest increases its angular velocity to 80 rad/s after 5 seconds. The total angular displacement is (1)
- A. 400 rad
 - B. 800 rad
 - C. 100 rad
 - D. 200 rad
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17. A satellite orbits Earth in a circular path with a period of 8 hours. If the orbital radius is increased to four times its original value, what will be the new orbital period? (1)
- A. 16 hours
 - B. 24 hours
 - C. 64 hours
 - D. 32 hours
-
18. A spacecraft hovers near a planet whose mass is twice that of Earth and radius 1.5 times larger than Earth's. The gravitational force experienced by the spacecraft compared to that near Earth is: (1)
- A. 8/9 of the gravitational force on Earth
 - B. 3/2 of the gravitational force on Earth
 - C. 4/9 of the gravitational force on Earth
 - D. Twice the gravitational force on Earth
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19. Students are asked to create a working model that applies Newton's third law in transportation. Which design BEST represents the principle clearly? (1)
- A. Create a solar-powered fan car that moves using a rotating blade
 - B. Design a gravity-powered cart that rolls down an inclined surface
 - C. Construct a rubber-band powered car that pulls itself forward mechanically
 - D. Build a balloon-powered vehicle that moves forward as air escapes backward
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20. According to Newton's second law in momentum form, if the net force on a body is zero, the momentum of the body: (1)
- A. Remains constant
 - B. Increases steadily
 - C. Decreases gradually
 - D. Changes direction
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21. Two blocks, A and B, each with mass of 10 kg, are connected by a light, inextensible string over a frictionless pulley. Block A rests on a 37° incline with coefficient of kinetic friction ($\mu_k = 0.2$) as shown in figure below. What is the acceleration of the system? (1)
- A. 1.20 m/s^2
 - B. 4.13 m/s^2
 - C. 2.52 m/s^2
 - D. 4.51 m/s^2
-
22. Two systems oscillate with negligible friction: a simple pendulum and a spring-mass system. Considering their energy transformations, which statement most accurately represents their energy behavior? (1)
- A. Energy remains entirely kinetic in both systems except at extreme positions
 - B. The spring system loses mechanical energy faster than the pendulum system
 - C. Only the pendulum exchanges kinetic energy while the spring stores constant energy
 - D. Both systems continuously exchange kinetic energy with potential energy during motion

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- 23.** A person leans against a wall and pushes hard, but the wall does not move. Later, the same person pushes a cart forward and it moves straight ahead. In which case is work done by the person? (1)
- A. Work is done in both cases because both involve a horizontal force
 - B. Work is done only when pushing the wall because more effort is used
 - C. No work is done in both cases because force is normal to displacement
 - D. Work is done only when pushing the cart because displacement occurs
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- 24.** A 0.2 kg tennis ball moving at $v = 40$ m/s is hit a wall and bounces back reversing its direction at $v = 40$ m/s. What is the magnitude of the impulse imparted to the ball? (1)
- A. 12 Ns
 - B. 16 Ns
 - C. 4 Ns
 - D. 0 Ns
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- 25.** A wedge of length 30 cm and thickness 6 cm is used to lift a load of 900 N with an effort of 300 N. What is its efficiency? (1)
- A. 50%
 - B. 60%
 - C. 75%
 - D. 90%
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- 26.** Which statement best distinguishes true weight from apparent weight? (1)
- A. True weight is measured in a fluid while apparent weight is measured in air
 - B. True weight changes with depth while apparent weight remains constant everywhere
 - C. True weight depends on liquid density while apparent weight does not depend on density
 - D. True weight is the gravitational force while apparent weight is the reduced weight in a fluid
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- 27.** Gauge pressure is best described as: (1)
- A. The total pressure acting on submerged surface
 - B. The pressure due to the density of a liquid only
 - C. The pressure measured relative to atmospheric pressure
 - D. The pressure measured relative to perfect vacuum condition
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- 28.** You are asked to design an experiment to demonstrate that pressure in a fluid increases with depth. Which setup would best illustrate this principle? (1)
- A. A vertical tube filled with water, having holes at different depths to compare leakage rates
 - B. A tilted glass partially filled with water, observing which side spills first as the water moves
 - C. A horizontal tube filled with water, open at one end, used to measure the flow speed of water
 - D. A u-shaped tube partially filled with water, with pressure measured only at the top using a gauge
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- 29.** A hydraulic lift is used to lift a car. The small piston has a radius of 5 cm and large piston has radius 50 cm. If a driver applies a force of 88 N to the small piston, what is the weight of the car the larger piston can support? (1)
- A. 880 N
 - B. 88 N
 - C. 88000 N
 - D. 8800 N
-
- 30.** A solid sphere floats fully submerged in a liquid. According to Archimedes' principle, which statement about the sphere's density is correct? (1)
- A. Sphere density = liquid density
 - B. Sphere density > liquid density
 - C. Sphere density < liquid density
 - D. Sphere density $\geq$ liquid density

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- 31.** A hose with diameter 4 cm delivers water at a speed of 2 m/s. The hose is connected to a nozzle of diameter 2 cm. What is the speed of water in the nozzle? (1)
- A. 16 m/s
B. 12 m/s
C. 4 m/s
D. 8 m/s
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- 32.** The process of using high pressure to kill or stop bacteria in food and increase its shelf life is called: (1)
- A. Sterilization
B. Pascalization
C. Fermentation
D. Pasteurization
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- 33.** Two substances, X and Y, each of mass 1 kg, undergo a temperature increase of 10 K. Substance X has specific heat 500 J/(kg·K) and substance Y has specific heat 800 J/(kg·K). Which statement correctly evaluates the heat required? (1)
- A. Neither substance requires any heat to raise its temperature
B. Substance Y requires more heat because its specific heat is larger than X
C. Substance X requires more heat because its specific heat is smaller than Y
D. Both require the same heat since their masses and temperature changes are equal
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- 34.** A steel bridge is 100 m long at 20°C. The linear expansion coefficient of steel is $1.2 \times 10^{-5} \text{K}^{-1}$. What length will it reach at 40°C? (1)
- A. 100.024 m
B. 100.04 m
C. 0.04 m
D. 0.024 m
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- 35.** During phase change, a substance is at temperature and pressure above its critical point. Which statement correctly describes its properties? (1)
- A. It exhibits both liquid-like and gas-like fluidity
B. It behaves as a liquid only, with a constant density
C. It solidifies immediately due to very high pressure
D. It behaves as a gas only, expanding to fill its container
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- 36.** You are tasked with redesigning the calorimetry experiment to minimize systematic heat loss. Which redesigned procedure best integrates multiple heat-loss prevention strategies? (1)
- A. Heat metal longer, transfer slowly, use less water, stir uncovered
B. Preheat calorimeter, transfer slowly, use more water, leave lid off
C. Insulate calorimeter, dry and transfer metal quickly, cover immediately, stir gently
D. Calibrate with known metal first, keep calorimeter open, transfer slowly, stir uncovered
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- 37.** Which statement correctly defines latent heat of fusion? (1)
- A. Energy required to change temperature of solid by one kelvin
B. Energy released when a solid is heated above its melting point
C. Heat required to change a liquid into gas at constant temperature
D. Total heat absorbed when a solid melts at constant temperature
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- 38.** A traffic light weighing 100 N hangs from vertical cables that are fastened to support. The upper cables make angles of 37° and 53° with the horizontal. What is the tension in T1 and T2 of cables respectively? (1)
- A. 60 N, 80 N
B. 80 N, 60 N
C. 35 N, 100 N
D. 100 N, 35 N

39. A copper wire has a resistance of $15\ \Omega$ and is connected to a 120 V supply. Due to heating, its resistance increases to $20\ \Omega$ while the voltage remains constant. What is the change in current? (1)

A. 8 A
B. 6 A
C. 3 A
D. 2 A

40. When identical resistors are connected in series and then reconnected in parallel to the same battery, how does the total current supplied by the battery compare? (1)

A. The total current becomes zero in parallel
B. The total current is the same in both arrangements
C. The total current is greater in parallel than in series
D. The total current is smaller in parallel than in series

41. Three charges $Q_1 = q$, $Q_2 = 3q$, and $Q_3 = -q$ are placed in a horizontal line. The charges are spaced equally by a distance d . What is the net force on the charge Q_2 ? (1)

A. $4kq^2/d^2$ to the right
B. $6kq^2/d^2$ to the right
C. $6kq^2/d^2$ to the left
D. Zero

42. An object 10 cm tall is placed 30 cm from a concave mirror of focal length 20 cm. The size of the image is (1)

A. 15 cm
B. 20 cm
C. 30 cm
D. 25 cm

43. The intensity of sound at 2 m from a point source is $8 \times 10^{-4}\ \text{W/m}^2$. What would be the intensity at 4 m from the same source, assuming no energy loss? (1)

A. $2 \times 10^{-4}\ \text{W/m}^2$
B. $4 \times 10^{-4}\ \text{W/m}^2$
C. $1 \times 10^{-3}\ \text{W/m}^2$
D. $8 \times 10^{-3}\ \text{W/m}^2$

44. A long straight wire carries a current of 10 A. What is the magnetic field strength at a point 2 cm from the wire? (1)

A. $1.0 \times 10^{-4}\ \text{T}$
B. $2.0 \times 10^{-5}\ \text{T}$
C. $5.0 \times 10^{-4}\ \text{T}$
D. $1.0 \times 10^{-5}\ \text{T}$

45. A student reverses the direction of current in a straight conductor and observes the compass near the wire. Which evaluation best explains the new observation? (1)

A. Compass shows no deflection because reversing current removes magnetic field
B. Compass rotates randomly because reversing current destroys circular magnetic field
C. Compass deflects opposite because reversing current reverses magnetic field direction
D. Compass deflects same because reversing current does not change magnetic field direction

46. A coil of 100 turns and area $0.01\ \text{m}^2$ rotates in a uniform magnetic field of 0.5 T. The angle between the coil's normal and the magnetic field changes from 30° to 60° in 0.1 s. What is the magnitude of induced electromotive force (EMF)? (1)

A. 1.33 V
B. 1.83 V
C. 2.00 V
D. 2.55 V

47. Which reason explains why transformers are commonly used in household power supply units? (1)

A. To directly convert alternating current into direct current
B. To efficiently change voltage levels for many appliances
C. To generate electricity using mechanical motion of devices
D. To safely store large amount of electrical energy for later use

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- 48.** Magnetic induction in graphics tablets allows the device to: (1)
- A. Track the pen's position without generating a signal
 - B. Store electrical energy to operate without being plugged in
 - C. Detect the pen's movement and convert it into digital lines
 - D. Produce heat on the screen to improve visibility for drawing
-
- 49.** Scientists attempt to date a 100,000-year-old rock sample using carbon-14. The measured amount of the isotope is extremely small and the result is unreliable. Which evaluation best explains why this dating method is inappropriate? (1)
- A. The isotope stops decaying after many years
 - B. The decay rate of the isotope changes in different rocks
 - C. The half-life of carbon-14 is too short for such old samples
 - D. The rock contains too many minerals that prevent measurement
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- 50.** A scientist compares two nuclear processes that release energy: Process I: Two light nuclei combine to form a heavier nucleus. Process II: A heavy nucleus splits into two medium-mass nuclei. Which explanation best analyzes why energy is released in both processes? (1)
- A. In both processes, the products have greater binding energy per nucleon, and the resulting mass defect appears as released energy
 - B. In both processes, neutrons are transferred or redistributed between nuclei, and this exchange directly produces the released energy
 - C. In both processes, extreme temperature or nuclear instability causes nuclei to change structure, and that condition itself produces the released energy
 - D. In both processes, unstable nuclei rearrange internally into more stable forms, and energy appears because nuclear forces are distributed during interaction
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- 51.** The main practical difference between forward and reverse bias is: (1)
- A. Both forward and reverse bias completely stop current
 - B. Forward bias blocks current, reverse bias allows current
 - C. Forward bias allows current, reverse bias blocks most current
 - D. Both forward and reverse bias allow the same amount of current
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- 52.** Two students compare ripple reduction in half-wave and full-wave rectifiers with identical capacitor filters. Which evaluation is correct? (1)
- A. Both rectifiers produce equally smooth DC when the capacitor is properly used
 - B. Half-wave rectifiers always produce smoother DC regardless of the capacitor values
 - C. Full-wave rectifiers cannot produce smoother DC without increasing the number of diodes
 - D. Full-wave rectifiers produce smoother DC because the capacitor charges twice in each cycle
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- 53.** A student is asked to design a two-input digital system that outputs 0 only when both inputs are 1 and outputs 1 for all other input combinations. Which of the following gates should the student select to correctly implement the design? (1)
- A. AND gate
 - B. OR gate
 - C. NOR gate
 - D. NAND gate

54. Robotic check-ups, needle-free devices, and electronic aspirin are electronic applications in:

(1)

- A. The aerospace and satellite control field
- B. The agricultural and crop monitoring field
- C. The medical and healthcare technology field
- D. The automobile and transport engineering field

— END OF EXAMINATION —

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